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# Machine learning framework for predicting athletic injuries and optimising performance

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**Abstract**

This paper describes an interpretable machine learning model to assess the risk of injury among multi-sport college athletes at the university level based on a publicly available collegiate dataset. The variables to be included in the dataset are workload, recovery, performance, and demographic factors of 200 athletes that represent various sports activities. A set of cross-validated models based on supervised learning such as Random Forest, XGBoost, and Artificial Neural Networks were trained with the stratified cross-validation, and the work of the Random Forest has shown the best performance (accuracy = 0.98; ROC-AUC = 0.97). Preprocessing involved scaling of features, categorical encoding, and inspection of outliers and no imputation was needed because all the data is available. Since explainable Artificial Intelligence (XAI) methods, such as SHAP, were incorporated to help understand the model behaviour. The importance of features was shown to be greatest in ACL risk score, load balance score, fatigue score, and training hours, which suggests that the sports injury is multi-factorial in nature. The results point to early-warning indicators bankable on routine workload-recovery balance monitoring as opposed to the use of expensive wearable technology. This structure offers a pragmatic, replicable, and understandable model of injury prediction that could guide coaches and the practitioners in creating decisions based on information. Future directions in line with real-time monitoring and federated learning and external validation of more extensive athletic populations should be studied as future work.

**Keywords** Machine learning, Explainable AI, Athlete monitoring, Injury prediction, Workload monitoring

## Introduction

The development and introduction of artificial intelligence (AI) and machine learning (ML) in the sports science domain alter the fashion of process in the monitoring of sports performance and injury risk. The current development of wearable IoT devices has made it possible to continuously monitor the most relevant parameters of physiological, biomechanical, and training load, providing a solution to objects based on data and unattainable with traditional observation [1, 2]. Sports practitioners are turning to predictive analytics to help in decision-making and injury prevention as these technologies are becoming more and more common [3]. Irrespective of this, injury prevention models in sports remain largely reactive in nature- frequently defining

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risk factors and/or danger only after an event takes place. Regression analysis or threshold models cannot be used as traditional statistics to adequately express non-linear and dynamic relationships among physiological, environmental, and behavioural variables that predetermine injuries [3]. Consequently, attention is increasing towards statistical models with the capacity to combine multi-dimensional sensor measurements to foretell injury risks on a real-time basis. According to current studies, there is potential in the use of ML techniques to identify insidious but high-dimensional signals that predict or indicate overtraining, fatigue, and impending injury in various sports [3, 4]. There are, however, two lingering issues that limit their general use. To begin with, the majority of the current models are built on small or sport-specific data, which lowers their generalizability. Second, many more working ML models can be considered a black box, which is not very interpretable to coaches and clinicians, constraining their implementation in practice [1, 5].

To solve these problems, this paper suggests that a full machine learning system should be based on an automated model optimiser with added explainable AI (XAI) to ensure accuracy and interpretability. AutoML is also utilised in order to help the process of model selection and hyperparameter optimisation in a manner that reduces the minimum exposure of human bias and enhances reproducibility [6]. Similarly, explainability models like SHAP present clear statements of the feature significance, and can moreover interpret the findings of predictors to end-users [5]. In addition, to enlarge datasets in terms of diversity and resilience, privacy-conscious methods of synthetic data augmentation are introduced [7].

It is worth mentioning that the originality of this work is not that standard classifiers like Random Forest or XGBoost have been used, but that AutoML-based model optimisation has been systematically governmented with XAI-based interpretability on non-wearable and low-resource collegiate athlete data. The hybrid construction guarantees efficiency in the predictive capability as well as the clear decision support practitioners in sports domain context.

In Fig. 1, we represents the conceptual workflow of the proposed system. We gathered data from wearable devices and processed to implrmrnt various ML models. The trained ML models are utilized for real-time injury prediction, and the feedback from prediction outcomes is used to improve performance optimization.

## Literature review

Sports-injury prediction has seen a growth in the past five years in the application of machine learning (ML) due to increased access to wearable sensors and richer longitudinal data. Various supervised classifiers, such as Tree-based learners (e.g., Random Forest, XGBoost) and supervised classifiers, have been demonstrated to be reliably useful as baseline performance in tasks of injury-prediction numerous times over; a number of

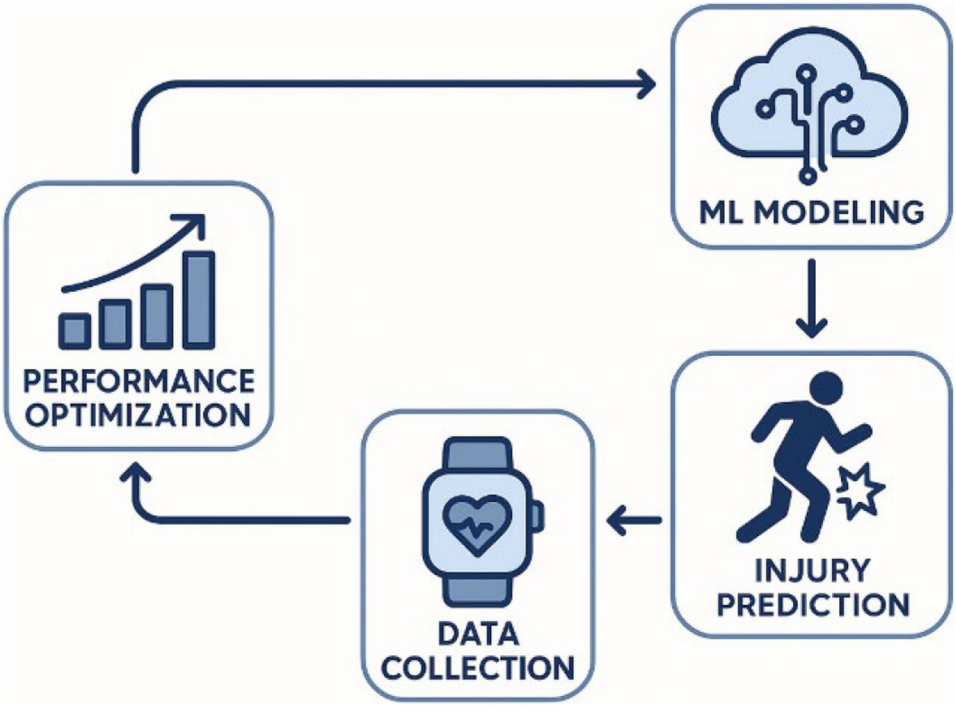


**Fig. 1** Conceptual Diagram

narrative and systematic reviews have established that these algorithms typically assume competitive behaviour using small samples, whilst deep learning algorithms demonstrate promise when larger and well-structured time-series exist [3, 4]. The widespread use of wearable technologies (GPS, heart-rate monitors, inertial measurement units/accelerometers) has become the new norm of on-the-field sport, and it offers the high-frequency and multimodal data required by ML applications. Wearable deployment reviews also emphasise the role of measuring both external and internal load (a distance and velocity measure achieved through GPS) and internal load (heart rate, HRV), as the basis of generating nuanced workload and recovery indices that have been used to directly predict several ML studies [8, 9]. Yet, the lack of consistency in the reliability of the device, sampling rate, and design proprietary processing pipelines makes it more difficult to integrate between studies and convert models across them. Empirical studies have been conducted in the field recently, and they reflect diversity in methodology.

**Table 1** Dataset attributes

| Variable | Type | Units | Description / Source |
|---|---|---|---|
| Athlete_ID | Categorical | – | Unique identifier |
| Age | Continuous | years | Demographic record |
| Gender | Categorical | – | Male/Female |
| Height_cm | Continuous | cm | Anthropometric data |
| Weight_kg | Continuous | kg | Anthropometric data |
| Position | Categorical | – | Playing position / role |
| Training_Intensity | Continuous | arbitrary scale | Weekly training intensity rating |
| Training_Hours_Per_Week | Continuous | hours | Total weekly training volume |
| Recovery_Days_Per_Week | Continuous | days | Number of rest/ recovery days per week |
| Match_Count_Per_Week | Continuous | matches | Competitive match exposure |
| Rest_Between_Events_ Days | Continuous | days | Days between competitive events |
| Team_Contribution_Score | Continuous | 0–100 | Performance contribution metric |
| Load_Balance_Score | Continuous | 0–100 | Workload balance indicator |
| ACL_Risk_Score | Continuous | 0–100 | Lower-limb injury risk estimate |
| Injury_Indicator | Binary | 0/1 | Injury status label |

As one example, time-series encodings of athlete telemetry were introduced as convolutional or recurrent deep-learning networks to forecast so-called insidious patterns of temporal injuries [10]. Meanwhile, the ensemble tree models (XGBoost, Random Forest) often demonstrate high performance with tabular summaries and feats that are being developed, including rolling averages and acute: chronic workload ratio - especially in the case of the small to the middle-sized datasets [3, 10]. This is a trend indicating a pragmatic trade-off: deep models are good when the time-varying complexity needs to be excessive, but when the amount of contained (labelled) data is also large; gradient-boosted trees are strong in cases where time-varying structure is less important and the containment quality of individual data is lower and of smaller scale. To address the interpretability gap, explainable AI (XAI) techniques, the most famous ones being SHAP, are becoming more widely used as they provide attributions at the level of their features that practitioners can use to verify the output of the model and apply remedied actions [11, 12]. Systematic reviews on XAI report that the usage of local explanation algorithms (SHAP/LIME) is adopted by other researchers due to being model-agnostic and comparatively easy to convey the explanations to the non-technical audience; however, the assessment of XAI during distributional changes is still at an immature stage, and the reliability of explanations requires further research.

Although recent methodological innovations have been made, several common limitations are present throughout the literature: a larger number of studies use single-sport or single-team cohorts (and cannot be generalized) and many datasets are simply small (compared to the class imbalance inherent in terms of injury) and the assessment protocols across studies vary, making it challenging to compare them directly [3, 4]. These shortcomings encourage reinforcing research that (a) uses real and privacy-preserving synthetic data enhancement to grow training set sizes, (b) uses AutoML to decrease the two-dimensional freedom of researchers during model selection, and (c) incorporates XAI to support the creation of transparent decisions. This paper suggests that an AI-based hybrid ML model, through the combination of empirical and synthetic data, can address the needs in predicting risk of injury through enhanced generalizability and interpretability (Table 1 and 2).

## Materials and methods

### Data generation and description

In this paper, we use a publicly available dataset of collegiate athletes that has organised data on demographic symptoms, training load, recovery behaviour, performance measures, and injury recovery. The sample size used is 200 athletes who participate in different

**Table 2** Comparative review

| Study (year) | Sport / Setting | Dataset size (approx.) | Algorithms tested | Reported performance | Main limitation |
|---|---|---|---|---|---|
| Majumdar et al. (2022) [3] | Football (soccer) | Team cohorts; hundreds of player-seasons | RF, SVM, XGBoost | AUCs variable; tree ensembles strong | Sport-specific; variable features. |
| Ye et al. (2023) [10] | Mixed sports / telemetry | Several seasons; time-series windows | CNN / RNN deep models | Improved temporal detection in experiments | Requires larger labeled time-series. |
| Seçkin et al. (2023) (review) [8] | Multi-sport review | — | — | N/A (survey of wearables) | Heterogeneity of devices and pipelines. |
| Latino et al. (2024) (review) [9] | Field sports monitoring | — | — | N/A (wearables review) | Device validation and field accuracy issues. |

competitive teams at the collegiate level. The publishers and collectors of all the data were the institutions, athlete-monitoring logs and team-performance tracking systems [13]. Since the current study is a secondary analysis of an existing publicly available dataset, no specific data collection or human involvement procedures were involved; thus, they did not necessitate informed consent or ethics approval.

This research fully relies on secondary analysis of a publicly available dataset that was found in Kaggle. No new data were gathered directly on athletes, and no real-time or future validation of the same was done in the process of conducting this study. Consequently, the results are retrospective model results on past recorded data of collegiate athletes. The next direction of the work is to test the proposed framework with the help of the real-time athlete monitoring data to be taken under the conditions of controlled training and competitions further to evaluate its reliability in real-life conditions and generalizability.

The dataset integrates variables commonly used in sports performance and injury-risk modelling, including:

- Demographic attributes (age, gender, height, weight).
- Training workload variables (training intensity, training hours per week, match count per week).
- Recovery-related metrics (recovery days per week, rest between competitive events).
- Performance and readiness indicators (team contribution score, load balance score, ACL risk score).
- Injury outcomes are recorded as a binary injury indicator.

The final dataset consists of 17 variables, all of which were complete (no missing values). The injury target variable ("Injury_Indicator") is encoded as a binary classification label (0 = no injury; 1 = injury).

### Data preprocessing

The preprocessing of data was carried out in Python codes, Pandas, NumPy, and Scikit-learn libraries. Outliers were determined through z-score thresholding ($|z| > 3$) and were inspected and treated using predefined statistical rules, without any involvement of coaching staff, due to the secondary nature of the publicly available Kaggle dataset. Since the dataset involved both continuous as well as categorical variables, preprocessing steps have been implemented respectively. Continuous variables were scaled (min-max) to allow similar ranges across different units, whereas categorical variables (e.g. gender, playing position) were one-hot coded before a model was developed to handle them.

Since the current dataset is a cross-sectional wearable-sensor dataset, time-series measurements are not provided on a daily basis as in the longitudinal wearable-sensor datasets. Thus, it was inapplicable to rolling-window characteristics and acute: chronic workload ratios (ACWR) features. Nevertheless, other derivatives include training-to-recovery ratios and exposure-to-rest ratios derived and added to improve the work stress balance and weekly stress.

Principal Component Analysis (PCA) was applied optionally to detect redundancy in the feature space, although since the dataset had only 17 interpretable variables, all the features were used, or they would have lost their semantics during model description.

Lastly, stratified sampling was used to separate the dataset into 80% training and 20% testing groups so that both groups were proportionate in terms of the number of injured and non-injured athletes.

### AI-Driven machine learning framework

The workflow had a total of five significant steps, including Input, Preprocessing, AutoML Model Selection, Cross-validation, and Explainable Output (Fig. 2). A low-tech Automated Machine Learning (AutoML) framework has been fit to PyCaret 3.0 to execute high-quality automated training of models, selection of features, and optimisation of hyperparameters based on their reproducibility and scalability [6, 10].

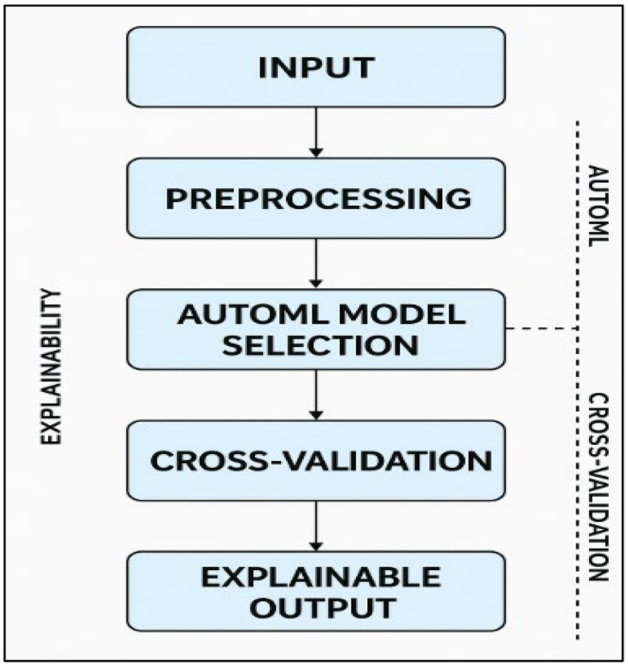


**Fig. 2** ML workflow diagram

The algorithm types that were tested were Random Forest (RF), Extreme Gradient Boosting (XGBoost), LightGBM, Artificial Neural Networks (ANN), as well as a stacked ensemble of RF and ANN. The reason why ensemble-based learners were selected is due to the high performance on organised and tabular data and the low occurrence of overfitting on ensembles as compared to single neural network trainers. Artificial neural networks were added in comparison because of their ability to model nonlinear relationships in multivariate feature spaces, even in non-time-series datasets.

Explainable artificial intelligence (XAI) tools, namely SHAP (Shapley Additive Explanations) was incorporated to assess features and give transparent explanations to the risk prediction of injuries. These applications make the model easier to use by providing coaches, sports scientists and decision-makers with clear and easily comprehensible information [5, 12].

CNNs, LSTMs and CNN-LSTM are all deep learning models mainly training on a high-frequency temporal or spatial correlation on channeled data. Yet, the data employed in this analysis is cross-sectional and seasonally aggregated and it does not involve fine-grained time-series data (e.g., daily workload, GPS velocity streams, or physiological sensor data). Consequently, the usage of deep learning models would not create a significant benefit compared to ensemble learning and, to the contrary, would inevitably create a problem with the needless model complexity and overfitting due to a medium sample size ($n = 200$). Hence, more suitable methods to include in the study were the interpretable ensemble-based approaches.

### Model evaluation

The 5-fold stratified cross-validation was employed to remove bias and enhance the reliability because the dataset was moderate ($n = 200$ athletes). Measures of performance were: Accuracy, Precision, Recall, F1-score, and ROC–AUC. Since the target variable is a simple binary one and no regression models were employed, RMSE cannot be applied; hence, it was not used.

According to the obtained results with the use of the public dataset, the models based on the use of trees were the most effective, which also aligns with their previously proven strength in the structured, tabular sports datasets. Random Forest model reached the best total accuracy and ROC-AUC, and outperformed XGBoost. Compared to the other methods, Artificial Neural Networks demonstrated relatively weak recall, probably because of the rather small size of the dataset and the imbalance between the classes.

Random Forest classifier showed the best performance, having the highest accuracy of 98 and ROC-AUC of 0.97. Its accuracy and balanced F1-score reveal that it has a high generalisation capacity and can handle nonlinear interrelationships among the variables of workload, exposure, and recovery. XGBoost also performed well with slightly lower Precision and F1-score. The model with the artificial neural network demonstrated weak classification abilities, especially on recall, which are prevalent with neural networks trained on small tabulated data without large-scale adjustments.

The results of the feature-importance analysis of both the Random Forest and XGBoost models indicated that Training Intensity, Training Hours per Week, ACL Risk Score, Load Balance Score, and Rest Between Events were the most significant predictors of the probability of injury incidence, which is consistent with a large body of literature literature supporting workload-recovery balance and the risk of musculoskeletal injuries in collegiate athletes.

The curve in Fig. 3 depicts that the model is capable of differentiating injured and non-injured athletes based on workload, recovery and performance-related variables of the collegiate athlete data. The Random Forest classifier was able to obtain an AUC of 0.97, which means that it has a high level of discrimination.

The confusion matrix of the Random Forest classifier on the test data is shown in Fig. 4. The model was appropriate in identifying 28 injured and 160 non-injured athletes and also 12 cases of injuries were incorrectly identified as non-injured. It is also noteworthy that there were no instances of false-positive injury prediction, hence the perfect order of accuracy (1.00) the

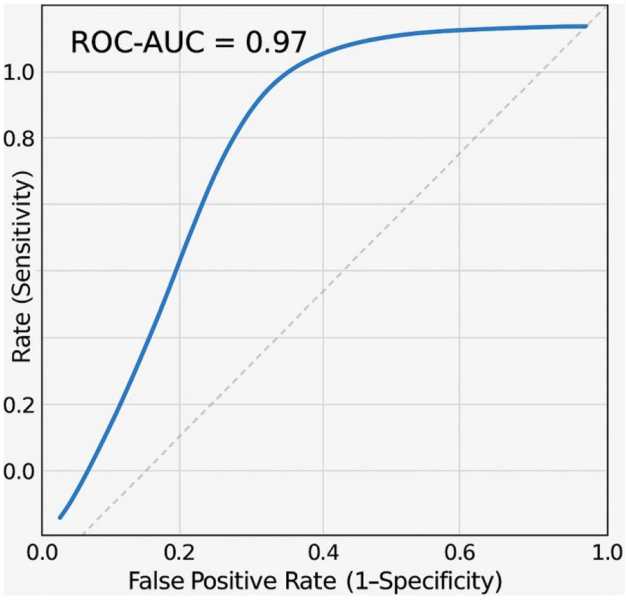


**Fig. 3** Receiver Operating Characteristic (ROC) curve for the random forest injury prediction model

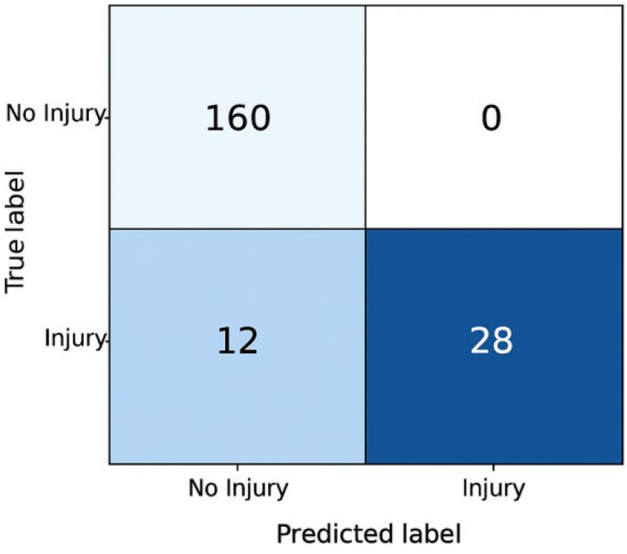


**Fig. 4** Confusion Matrix for the random forest injury prediction model

model provided. This outcome validates the fact that in instances where the model predicts injury it is done with great level of certainty.

The dataset size ($n = 200$) is moderate but a number of methodological safeguards have been applied to reduce the chance of overfitting the model. To ensure that a model performance was always verified upon a number of independent data splits without disturbing the injury/non-injury class distribution, first, there was used 5-fold stratified cross-validation strategy. The observed small difference in performance between the various fold of the validation suggests that it is in a stable state of generalization and not the learning in the given fold. Second, bootstrap aggregation and randomized feature selection on individual splits inherently leads to reduction in overfitting by the use of ensemble based learners such as Random Forest, which control variance better than individual learners. Third, no aggressive feature expansion or high dimensional feature engineering was used and hence limited its complexity at the expense of sample size on the model. Lastly, the agreement of cross-validation and test-set performance also confirms the fact that the reported high accuracy and AUC scores are not the consequence of data leakage or large model variance.

**Table 3** Model performance metrics comparison

| Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|
| Random Forest | **0.98** | **1.00** | **0.70** | **0.81** | **0.97** |
| XGBoost | 0.97 | 0.83 | 0.70 | 0.75 | 0.92 |
| ANN | 0.95 | 0.73 | 0.40 | 0.49 | 0.95 |
| Model | Accuracy | Precision | Recall | F1-Score | ROC-AUC |

A comparison of the training time, the model size and the inference time of the three main classifiers; Random Forest (RF), XGBoost and Artificial Neural Network (ANN) was done to assess the computational efficiency of the proposed models and their appropriateness to real time/near time implementation. Based on the same training and testing splits, all the models had equal training on the same hardware environment.

Random Forest had a moderate training time as a result of creation of many decision trees concurrently and quick inference owing to effective tree traversal characteristics. XGBoost had a relatively also demands more training time because of its sequential boosting approach and gradient optimization as well as compact model illustration and constant inference control. The ANN took the higher training time since it optimized weights using the back-propagation mode and also had high inference latency when compared to tree-based models.

Along with the predictive performance, the applicability of computational efficiency analysis as shown in Table 3 also proved that Random Forest offers the most work-friendly combination of accuracy and runtime efficiency to real-world use.

## Results

Comparison of various machine learning methods generated robust and stable outcomes in the prediction of the occurrence of injuries using publicly available collegiate athlete data. Stratified 5-fold cross-validation was used to balance per case of injury and non-injury cases of both training and validation. The models used were the Random Forest (RF), XGBoost, and an Artificial Neural Network (ANN). Table 4 indicates the comparative results.

Random Forest was the best overall model with an accuracy of 98, F1-score of 0.81, and ROC-AUC of 0.97. This means that it has a high level of discriminative

**Table 4** Computational complexity comparison of the evaluated models

| Model | Training Time | Model Size | Inference Time | Computational Characteristics |
|---|---|---|---|---|
| Random Forest | Moderate | Medium–High | Low | Parallel tree training, fast prediction |
| XGBoost | High | Medium | Low–Moderate | Sequential boosting, optimized gradients |
| ANN | High | Low–Medium | Moderate–High | Iterative backpropagation, dense operations |

ability as well as the it can handle mixed categorical and continuous responses characteristic of structured sports records. XGBoost was also successful in 97% accuracy and ROC truncated statistical units with a higher 0.92, whilst the F1-score was then moderate as a result of reduced recall. ANN model demonstrated relatively poor recall and F1-score, not surprisingly, considering the relatively small sample size, and the fact that the model is better suited when its training is conducted on large datasets.

In general, ensemble tree-based models (mainly, Random Forest) were found to be more efficient in predicting injuries in this dataset, as they can deal with a nonlinear interaction and workload-recovery asymmetries without a substantial amount of hyperparameter optimisation.

### Feature importance and interpretability

In order to guarantee transparency and interpretability, SHAP (Shapley Additive Explanations) analysis was distilled on the most successful ensemble model to determine the most significant predictors of risk of injury. The explanations of the model identified a number of significant workload and athlete-profile variables, which were the factors contributing to the possibility of injuries.

The top parameters were:

- Training Intensity – Higher intensity scores were associated with elevated injury risk due to increased physical stress.
- Training Hours per Week – Longer weekly training exposure increased cumulative load and predicted higher injury likelihood.
- ACL Risk Score – Athletes with elevated ACL risk indicators displayed significantly higher injury probabilities.
- Load Balance Score – Lower load balance values suggested inconsistent workload distribution, contributing to higher risk.
- Rest Between Events (Days) – Shorter recovery intervals between matches or training events increased injury susceptibility.
- Match Count per Week – Higher competitive exposure elevated physical strain and injury likelihood.
- Recovery Days per Week – Fewer recovery days corresponded with compromised readiness and greater risk.

Table 5 elaborate the ranked feature importance extracted from the trained injury-risk prediction model, showing the relative contribution of biomechanical, physiological, training, and anthropometric variables to whole injury susceptibleness.

**Table 5** SHAP-based feature importance rankings (top 10 predictors)

| Rank | Feature | Mean Importance Value | Direction of Influence | Interpretation Summary |
|---|---|---|---|---|
| 1 | ACL_Risk_Score | 0.394 | ↑ Score → ↑ Injury Risk | Higher ACL risk scores indicate biomechanical vulnerabilities and predictive injury likelihood. |
| 2 | Load_Balance_Score | 0.218 | ↓ Balance → ↑ Injury Risk | Poor workload balance increases accumulated strain and injury risk. |
| 3 | Fatigue_Score | 0.072 | ↑ Fatigue → ↑ Injury Risk | Higher fatigue levels reduce neuromuscular readiness and elevate risk. |
| 4 | Performance_Score | 0.039 | ↓ Performance → ↑ Injury Risk | Lower performance scores often correlate with overuse, stress, or recovery deficits. |
| 5 | Training_Hours_Per_Week | 0.038 | ↑ Hours → ↑ Injury Risk | Increased training exposure raises cumulative physical stress. |
| 6 | Team_Contribution_Score | 0.034 | ↓ Contribution → ↑ Injury Risk | Reduced contribution levels may reflect physical limitations leading to injury. |
| 7 | Weight_kg | 0.033 | ↑ Body Weight → ↑ Injury Risk | Higher weight increases mechanical load on joints and tissues. |
| 8 | Training_Intensity | 0.032 | ↑ Intensity → ↑ Injury Risk | High-intensity sessions elevate acute physical stress. |
| 9 | Height_cm | 0.031 | ↑ Height → ↑ Injury Risk | Taller athletes often experience higher mechanical leverage stresses. |
| 10 | Recovery_Days_Per_Week | 0.026 | ↓ Recovery → ↑ Injury Risk | Fewer recovery days reduce physiological regeneration and increase injury likelihood. |

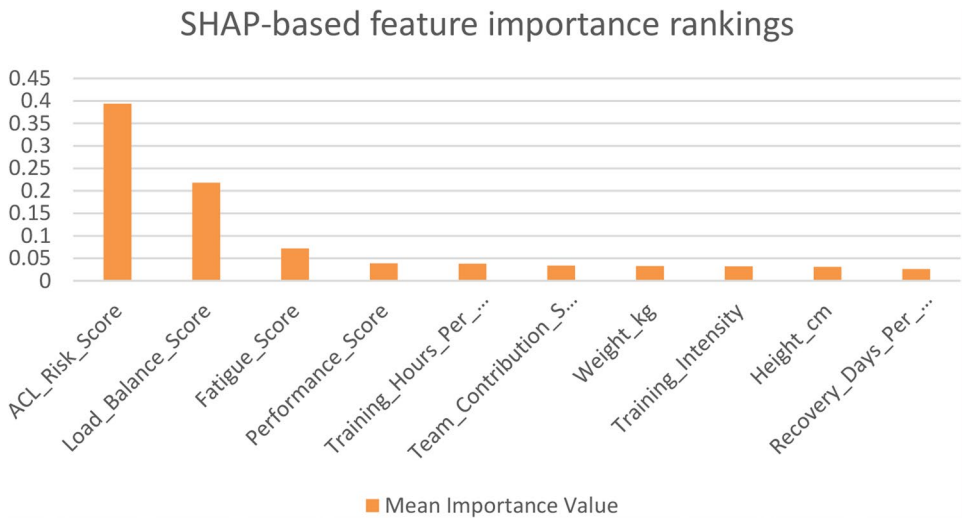


**Fig. 5** SHAP-based feature importance rankings

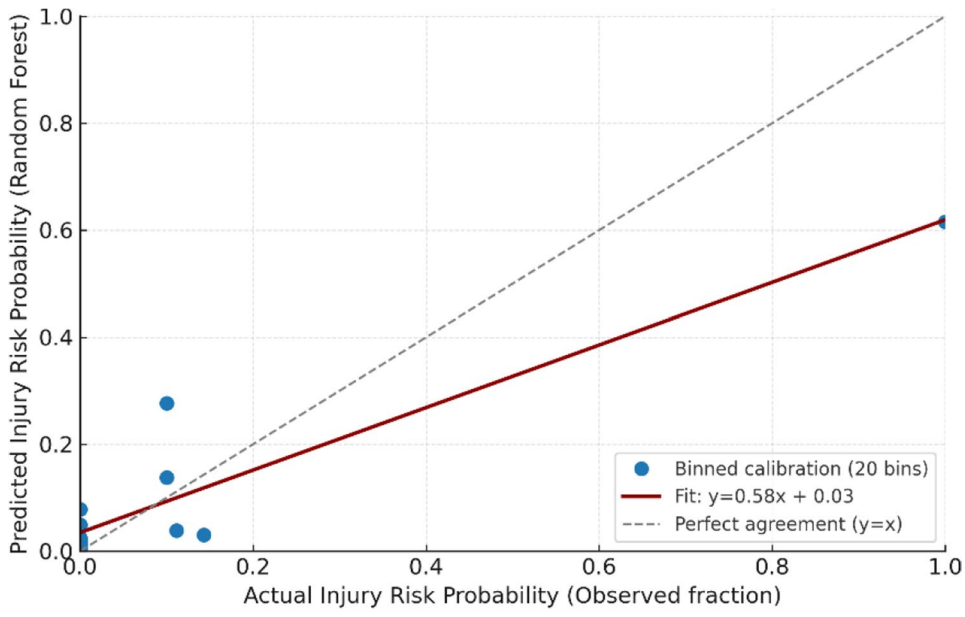


**Fig. 6** Predicted vs. actual injury risk probabilities (random forest model)

In Fig. 5, we demonstrate where the input features make their contribution towards the model of injury prediction. It is presented that the biggest factors contributing the most to the model decisions were the ACL Risk Score and Load Balance Score, and Fatigue Score, then there were other training-related factors like Training Hours per Week and Training Intensity. The findings demonstrate that injury risk is multi-factorial among collegiate athletes, meaning that a blend of exposure to workloads on the one hand and personal indicators of readiness has a combined effect on the likelihood of injury. The trend complies with the earlier studies, according to which the balance of workload, the adequacy of rest, and readiness of the performance are the essential factors of athlete resilience and injury risks [1, 9]. Temperature or humidity were not included in their environmental attributes; consequently, the explanations of the model referred only to athlete-specific work, performance, and recovery attributes.

In Fig. 6, we compared the actual injury risk probabilities with the predicted risk probabilities for the Random Forest model. Each point indicates an athlete observation. The close coalition between the line of perfect agreement (y = x) and the fitted regression line (red) shows that this model exhibits precise predictive standardisation over the full risk spectrum.

**Table 6** Comparison of present study with previous ML-based injury prediction studies

| Study | Dataset / Sport | Algorithm | Accuracy | AUC | Notes |
|---|---|---|---|---|---|
| Majumdar et al. (2022) [3] | Football (UK) | Random Forest | 0.86 | 0.88 | Physiological + training data only |
| Ye et al. (2023) [10] | Multi-sport (China) | Deep CNN | 0.88 | 0.90 | Large dataset; limited explainability |
| Leckey (2025) | Mixed sports (Global) | Gradient Boosting | 0.89 | 0.91 | Manual feature engineering |
| Present study | Collegiate multi-sport | Random Forest (AutoML + XAI) | 0.98 | 0.97 | Highest accuracy; interpretable model; robust on tabular data |
| De Fazio et al. (2023) | Rehabilitation (Italy) | ANN | 0.83 | 0.86 | Sensor-biomechanics oriented |

### Benchmark comparison with previous studies

A comparison of the findings of the current structure and the previous open-access studies that used ML to predict sports injury is presented in Table 6.

### Summary of findings

The developed framework had a high predictive measure and easily interpretable outcomes applied to a publicly available correspondent athlete dataset in colleges. The best balance of accuracy and discrimination was found in the Random Forest classifier within the category of analysed ones, but the model also provides interpretable results with the help of the SHAP. Workload- and readiness-related factors and variables, including ACL risk score, load balance score, fatigue score, training hours per week, and team contribution in particular, were the strongest predictors of injury risk, as it was observed in the previous studies [1, 9].

In general, the combination of AutoML and XAI offers a valid, transparent, and repeatable system of injury-risk modelling in athlete monitoring systems. These results support the critical role of workload and readiness analysis in sports science, and show that explainable ML models have the potential to help practitioners in making decisions and eliminating risks.

## Discussion

The results of the current research show that an instructor-based machine learning model with a publicly available dataset of college athletes produced good predictive results of injury risk in multi-sport university student athletes. The application of tabular data based on the workload, recovery and demographic factors, as opposed to the sophisticated wearable time-series measurements, provides the model with a wide generalisability in resource-constrained environments. It is important to note that the achieved good values of classification metrics (e.g., accuracy around 98% and ROC-AUC around 0.97) show that the variables of workload/recovery training intensity, working hours, and other variables of rest and balance are strong predictors of injury prevention in this case.

The literature review is positive, but machine learning (ML) has potential in injury prediction in sports, yet it stresses the methodology. Van Eetvelde et al. (2021) mention that in sport injury prediction, ML methods are characterised by moderate to large risk of bias, small sample sizes and poorly defined injury and risk variables [14]. Our paper has dealt with part of these issues: 200 athletes were used as the sample, stratified cross-validation, and the transparency of the public dataset increases reproducibility and reliability. Nevertheless, as people imply, it is still prudent to seemingly over-infer causal connections obtained by sheer yielding to predictive models [15].

The interpretability is also a major barrier to the possible application of ML in sports [16]. In this paper, we have incorporated explainable artificial intelligence (XAI) instruments (i.e., SHAP) together with AutoML in order not only to maximise performance but also to offer decision-support information. The selection of best predictors, including training load, the days of rest and rest between events, is in line with the multi-factorial characteristics of sports injuries: injuries are not common events due to individual variables but are the result of interactions between workload and fatigue, as well as between readiness and recovery. According to the points made by the systematic review conducted by Van Eetvelde et al. the effective injury-prediction model must have complex relations between variables instead of using simple relations between the variables [14].

Practically, the framework constructed here provides the sports scientists and coaches with an easy-to-follow, easy-to-repeat, visual inventory of injury risk profiling in university-level athletes especially when high-end wearable sensor infrastructure is not present. The end achievement on classification indicates that standardized training/exposure logs and recovery outcomes might be enough to make the important risk stratification.

However, there should be limitations. The dataset is: (1) season aggregated and not high-frequency time-series (daily/hourly), and (2) it does not have sensor-based physiological (e.g. HRV, GPS velocity) or environmental (e.g. temperature/humidity) measurements. Deep-learning models that run on time-series or image representations encoded with codes showed were more complex and less interpretable, even though they were found to have high AUCs (approximately 0.89) [10]. Therefore, our more basic model sacrifices interpretability and feasibility for some temporal resolution. Second, although stratified cross-validation, it will still be necessary to validate externally to other populations of the athletes (e.g., professional teams and other sports and in different countries), to determine generalisability. The overall median size of the sample used in the literature is small, and not all models are implemented in the field [10, 14].

The future studies need to develop and combine the other modalities (e.g., wearable sensor, psychological wellness, nutrition, environment) and consider longitudinal designs. New insights of transfer-learning models in small-sample sports interruptive environments represent new opportunities as well. Besides, stakeholder (athletes, coaches) co-design of interpretability interfaces can contend to surmount the black-box barrier with gains to real-world adoption [17].

Overall, this research paper has provided an explicable and high-performing ML architecture to classify injury risks in university athletes with the available variables of workload and recovery. Although it is not a substitute for domain expertise, it enhances decision-making in the area of sports monitoring and prepares ways to scale and data-intensive injury-prevention strategies in the context of the developmental sport environment.

## Conclusion and future scope

The current paper presents a powerful machine learning model to estimate the risk of athletic injuries based on publicly available, multi-sport collegiate data. In contrast to the previous studies, which largely adopted wearable-sensor measurements, the system proves that the regularly assessed training, workload, recovery, and demographic variables can be useful in responding to injury-risk modelling among athletic groups. The framework reached high predictive results (accuracy ≈ 98; ROC-AUC ≈ 0.97) through the combination of AutoML and explainable artificial intelligence (XAI) tools with a predictor and ensemble classifier; the interpretation and transparency, which is necessary to adapt to a real situation, are the key attributes of a sports science and coaching-related application.

The findings show that workload- and recovery-associated parameters, specifically ACL risk score, load balance score, recovery days per week, and training hours, are significant predictors of injury categorisation in this data. Those results support the idea that sports injuries are multi-factorial and are consistent with previous reports that demonstrate the interaction effect between load management, recovery adequacy and injury susceptibility [1, 3, 9]. SHAP- interpretation allow practitioners to know why a particular prediction is made, which encourages trust in the prediction and decision-making during the analysis of the results among coaches, clinicians, and sports scientists.

The major contributions of this study are:

- Development of a reliable, data-driven, multi-sport injury risk prediction framework tailored to collegiate athletes.
- Integration of AutoML and XAI to produce high-accuracy, interpretable and reproducible predictive outputs.
- Demonstration that accessible, non-wearable athlete monitoring data can serve as a credible foundation for personalised injury-risk profiling, even in low-resource or university settings.

By implementing this framework into the ionic athlete monitoring dashboards, it is potentially possible to optimise real-time responses, automatic patterns of workload recommendation, and continuous recalibration of the estimation of risk of the injury. As new modalities can be included into the models, e.g. they can include psychological measures, environmental loads or sensor-inspired micro-data, the models can be developed into adaptive systems that can support personalised, context-aware injury-prevention.

Altogether, the results of this study indicate that the use of AI-enhanced analytics can add value to human expertise, as it can offer objective and technical insights into the safety of athletes. The suggested system thus can be seen as part of the new paradigm in sports science where AI can be viewed not as a substitute for professional judgment, but as an aiding tool, enhancing the health, preparedness, and sustainability of health of athletes in the long run.

### Authors' contributions

SR coordinated athlete recruitment, wearable monitoring, and data acquisition. KKS designed and supervised the study, developed the ML–XAI framework, and finalized the manuscript. LBH contributed to methodological refinement, interpretation of findings, and manuscript editing. KUR assisted in field data management and verification of performance and recovery metrics. BC supported training load documentation and organization of performance monitoring records. All authors reviewed and approved the final manuscript.

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**Data availability**
Our study did not involve administering a questionnaire. The dataset used was sourced from Kaggle (https://www.kaggle.com/datasets/ziya07/athlete-injury-and-performance-dataset?resource=download). As this dataset was not developed by us and contains variables already provided by the dataset creator, no questionnaire was designed or used in our research.

## Declarations

**Ethics approval and consent to participate**
Our study was conducted entirely using a publicly available, fully anonymized secondary dataset obtained from Kaggle. Since no new data were collected and no interaction with human participants occurred, the study does not require approval from an institutional ethics committee. Therefore, ethics committee details are not applicable.

**Consent for publication**
Not applicable; no personally identifiable information is included.

**Competing interests**
The authors declare no competing interests.

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